# R112/A_collapsed | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 20 | update 57600 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) |
|---|---|---|---|---|---|---|---|---|
| VAL ALL | 71.0605 | 66.2037 | 11.417398 | 11.427978 | 11.371692 | -0.1878 | +0.3999 | 0.4106 |
| TEST ALL | 71.2840 | 66.1802 | 11.374503 | 11.385299 | 11.329932 | -0.1909 | +0.3923 | 0.4003 |

| Dataset | n | NSS Top1 (%) | R112 Top1 (%) | NSS cost | R112 cost |
|---|---|---|---|---|---|
| TSP TEST | 4500 | 45.888889 | 46.955556 | 7.970400719 | 7.972858822 |
| CVRP TEST | 4500 | 45.977778 | 48.066667 | 18.004802368 | 18.012467872 |
| TSPLIB | 49 | 20.408163 | 32.653061 | 8.217907074 | 8.196546667 |
| CVRPLIB | 100 | 70.000000 | 64.000000 | 67.101111717 | 67.242448912 |

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS: saved exact4500/native LIB references only; MISSING entries are not inferred or retrained.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

# R112/A_collapsed | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) | TRAIN SBS |
|---|---|---|---|---|---|---|---|---|---|
| ALL | 71.0605 | 66.2037 | 11.417398 | 11.427978 | 11.371692 | -0.1878 | +0.3999 | 0.4106 | per-problem TRAIN SBS |
| TSP | 45.9333 | 33.1111 | 8.027187 | 8.045354 | 7.990533 | -0.2258 | +0.4587 | 0.4469 | T2T |
| CVRP | 47.1333 | 18.7111 | 18.163369 | 18.269377 | 18.064087 | -0.5803 | +0.5496 | 0.7316 | RELD_CVRP |
| ATSP | 77.1778 | 47.4889 | 1.638411 | 1.676905 | 1.632471 | -2.2955 | +0.3639 | 0.3610 | UNICO_MatPOENet |
| OVRP | 41.0889 | 35.7556 | 7.462179 | 7.469579 | 7.415777 | -0.0991 | +0.6257 | 0.6395 | MoSES_RF |
| VRPB | 80.8222 | 80.8444 | 9.356876 | 9.354876 | 9.318750 | +0.0214 | +0.4091 | 0.4177 | reld_mt |
| VRPL | 37.5778 | 33.0889 | 12.432051 | 12.437959 | 12.383444 | -0.0475 | +0.3925 | 0.3993 | MoSES_RF |
| VRPTW | 80.1778 | 79.9556 | 18.817987 | 18.819431 | 18.766139 | -0.0077 | +0.2763 | 0.2911 | reld_mt |
| OVRPTW | 83.7556 | 83.7778 | 10.143410 | 10.143730 | 10.116859 | -0.0032 | +0.2624 | 0.2604 | reld_mt |
| OVRPB | 88.5333 | 88.5111 | 6.357313 | 6.357254 | 6.341078 | +0.0009 | +0.2560 | 0.2650 | reld_mt |
| OVRPL | 41.2000 | 35.9111 | 7.438125 | 7.445709 | 7.390416 | -0.1019 | +0.6455 | 0.6617 | MoSES_RF |
| VRPBL | 79.0000 | 78.5111 | 9.371116 | 9.371523 | 9.328366 | -0.0043 | +0.4583 | 0.4658 | reld_mt |
| VRPBTW | 78.6444 | 78.4222 | 19.947534 | 19.950430 | 19.851084 | -0.0145 | +0.4859 | 0.4631 | reld_mt |
| VRPLTW | 78.9333 | 78.7556 | 18.764927 | 18.765871 | 18.707843 | -0.0050 | +0.3051 | 0.3188 | reld_mt |
| OVRPBL | 89.2667 | 89.2000 | 6.369687 | 6.370093 | 6.352824 | -0.0064 | +0.2654 | 0.2756 | reld_mt |
| OVRPBTW | 83.5111 | 83.4444 | 10.660699 | 10.661098 | 10.623245 | -0.0037 | +0.3526 | 0.3267 | reld_mt |
| OVRPLTW | 83.5111 | 83.4889 | 10.157368 | 10.157423 | 10.129818 | -0.0005 | +0.2720 | 0.2765 | reld_mt |
| VRPBLTW | 79.5333 | 79.4000 | 19.794218 | 19.797044 | 19.706420 | -0.0143 | +0.4455 | 0.4380 | reld_mt |
| OVRPBLTW | 83.2889 | 83.2889 | 10.610700 | 10.609955 | 10.571302 | +0.0070 | +0.3727 | 0.3523 | reld_mt |

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

# R112/A_collapsed | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) | TRAIN SBS |
|---|---|---|---|---|---|---|---|---|---|
| ALL | 71.2840 | 66.1802 | 11.374503 | 11.385299 | 11.329932 | -0.1909 | +0.3923 | 0.4003 | per-problem TRAIN SBS |
| TSP | 46.9556 | 34.5333 | 7.972859 | 7.989621 | 7.937593 | -0.2098 | +0.4443 | 0.4347 | T2T |
| CVRP | 48.0667 | 17.9333 | 18.012468 | 18.124983 | 17.918509 | -0.6208 | +0.5244 | 0.6892 | RELD_CVRP |
| ATSP | 77.0222 | 47.2444 | 1.635269 | 1.674207 | 1.629441 | -2.3258 | +0.3576 | 0.3569 | UNICO_MatPOENet |
| OVRP | 40.6444 | 33.6444 | 7.445991 | 7.455750 | 7.400487 | -0.1309 | +0.6149 | 0.6207 | MoSES_RF |
| VRPB | 80.2444 | 80.4000 | 9.355491 | 9.351810 | 9.315298 | +0.0394 | +0.4315 | 0.4319 | reld_mt |
| VRPL | 37.8444 | 32.2667 | 12.349429 | 12.356138 | 12.300958 | -0.0543 | +0.3940 | 0.4004 | MoSES_RF |
| VRPTW | 81.0889 | 80.8444 | 18.688367 | 18.690545 | 18.637408 | -0.0117 | +0.2734 | 0.2845 | reld_mt |
| OVRPTW | 83.8889 | 83.9333 | 10.093007 | 10.093063 | 10.068135 | -0.0006 | +0.2470 | 0.2383 | reld_mt |
| OVRPB | 88.1111 | 88.1333 | 6.371938 | 6.370826 | 6.352225 | +0.0175 | +0.3103 | 0.3221 | reld_mt |
| OVRPL | 41.6889 | 34.6000 | 7.464033 | 7.472542 | 7.418033 | -0.1139 | +0.6201 | 0.6310 | MoSES_RF |
| VRPBL | 79.2889 | 79.4222 | 9.396092 | 9.396379 | 9.355967 | -0.0031 | +0.4289 | 0.4274 | reld_mt |
| VRPBTW | 78.7778 | 78.7556 | 19.664989 | 19.665725 | 19.577666 | -0.0037 | +0.4460 | 0.4410 | reld_mt |
| VRPLTW | 79.2444 | 79.2444 | 18.802867 | 18.803136 | 18.745143 | -0.0014 | +0.3079 | 0.3234 | reld_mt |
| OVRPBL | 89.2000 | 89.2889 | 6.375536 | 6.375741 | 6.359310 | -0.0032 | +0.2551 | 0.2674 | reld_mt |
| OVRPBTW | 84.9556 | 84.8222 | 10.658937 | 10.659591 | 10.624748 | -0.0061 | +0.3218 | 0.2943 | reld_mt |
| OVRPLTW | 82.8889 | 82.8667 | 10.085998 | 10.086112 | 10.059001 | -0.0011 | +0.2684 | 0.2699 | reld_mt |
| VRPBLTW | 79.6889 | 79.7333 | 19.705507 | 19.707068 | 19.613284 | -0.0079 | +0.4702 | 0.4493 | reld_mt |
| OVRPBLTW | 83.5111 | 83.5778 | 10.662269 | 10.662149 | 10.625574 | +0.0011 | +0.3453 | 0.3234 | reld_mt |

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

# R112/A_collapsed | LIB and locked provenance

Current R112 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

| Split / Problem | Top1 (%) | SBS Top1 (%) | mean_cost | SBS cost | Oracle cost | vs_SBS (%) | vs_Oracle (%) | actual regret (%) |
|---|---|---|---|---|---|---|---|---|
| TSPLIB | 32.6531 | 10.2041 | 8.196547 | 8.240435 | 8.142167 | -0.5326 | +0.6679 | 0.6864 |
| CVRPLIB | 64.0000 | 45.0000 | 67.242449 | 67.245364 | 66.945093 | -0.0043 | +0.4442 | 0.5385 |

SHA256: 4bf692f36fde2f3f7ce461ccc96a01b2bd254e606c9ea40c7f5e3aea9c44ae3e

Unique selection: strict maximum full18 VAL macro_top1; exact Top1 ties: minimum macro_actual_regret_pct; exact double ties: first checkpoint

Inference: eval mode, original R77 score-difference maximin; stable ascending-global-solver-ID ties

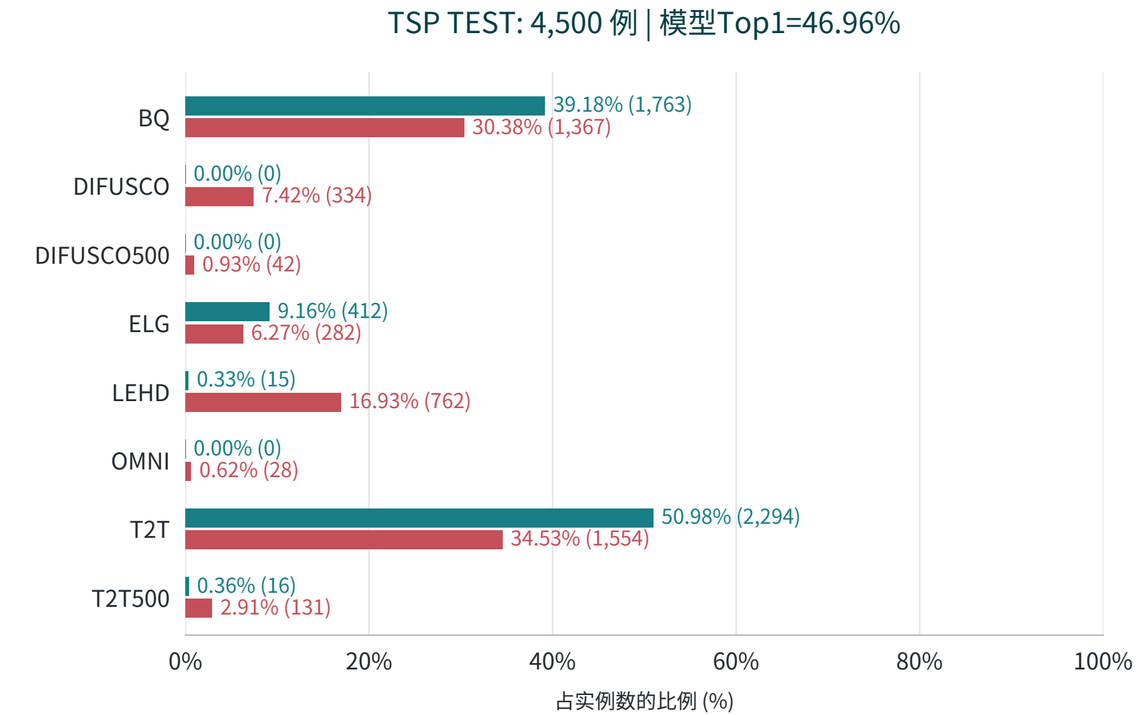
Normalization: new81000 TRAIN only; no TEST/LIB refit.

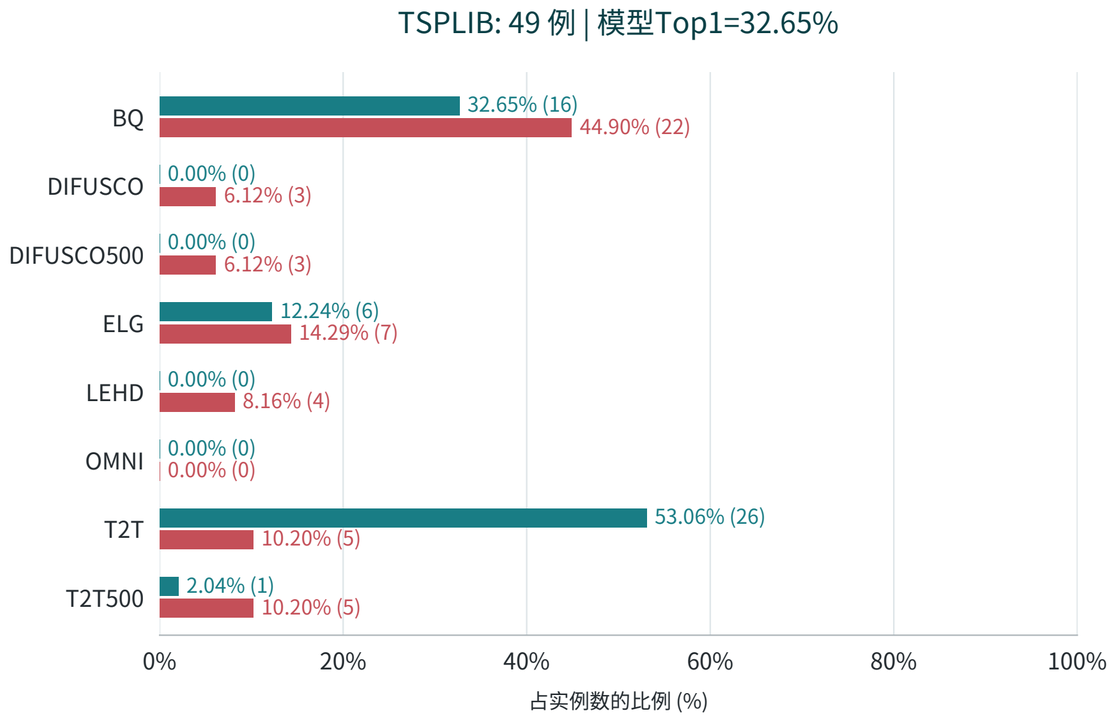
# R112/A_collapsed | TSP | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

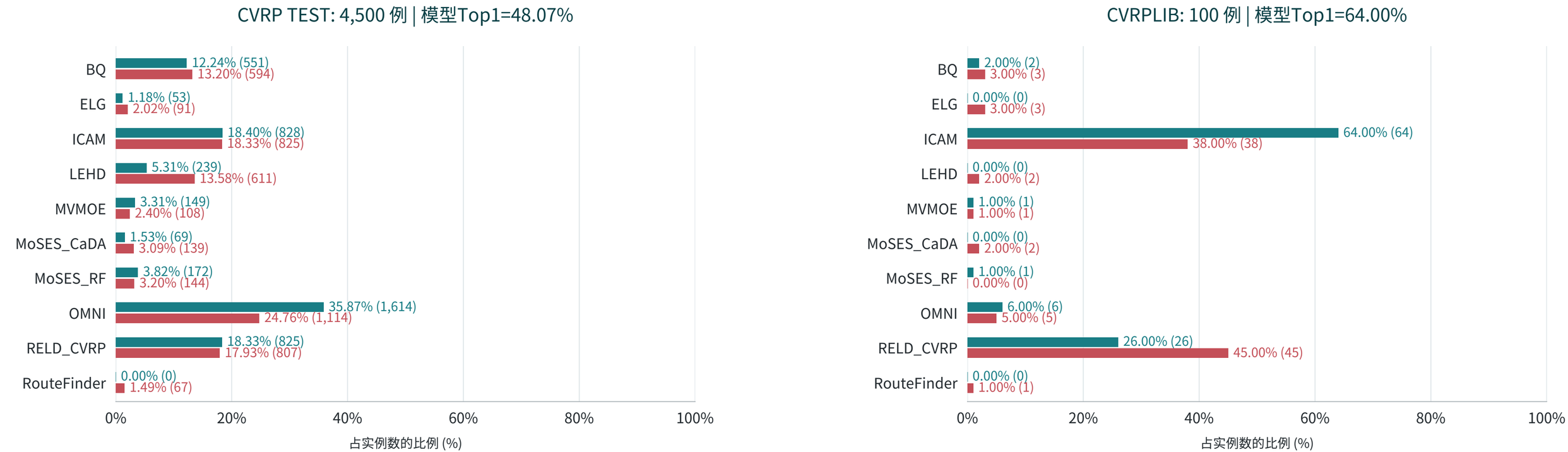
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | CVRP | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

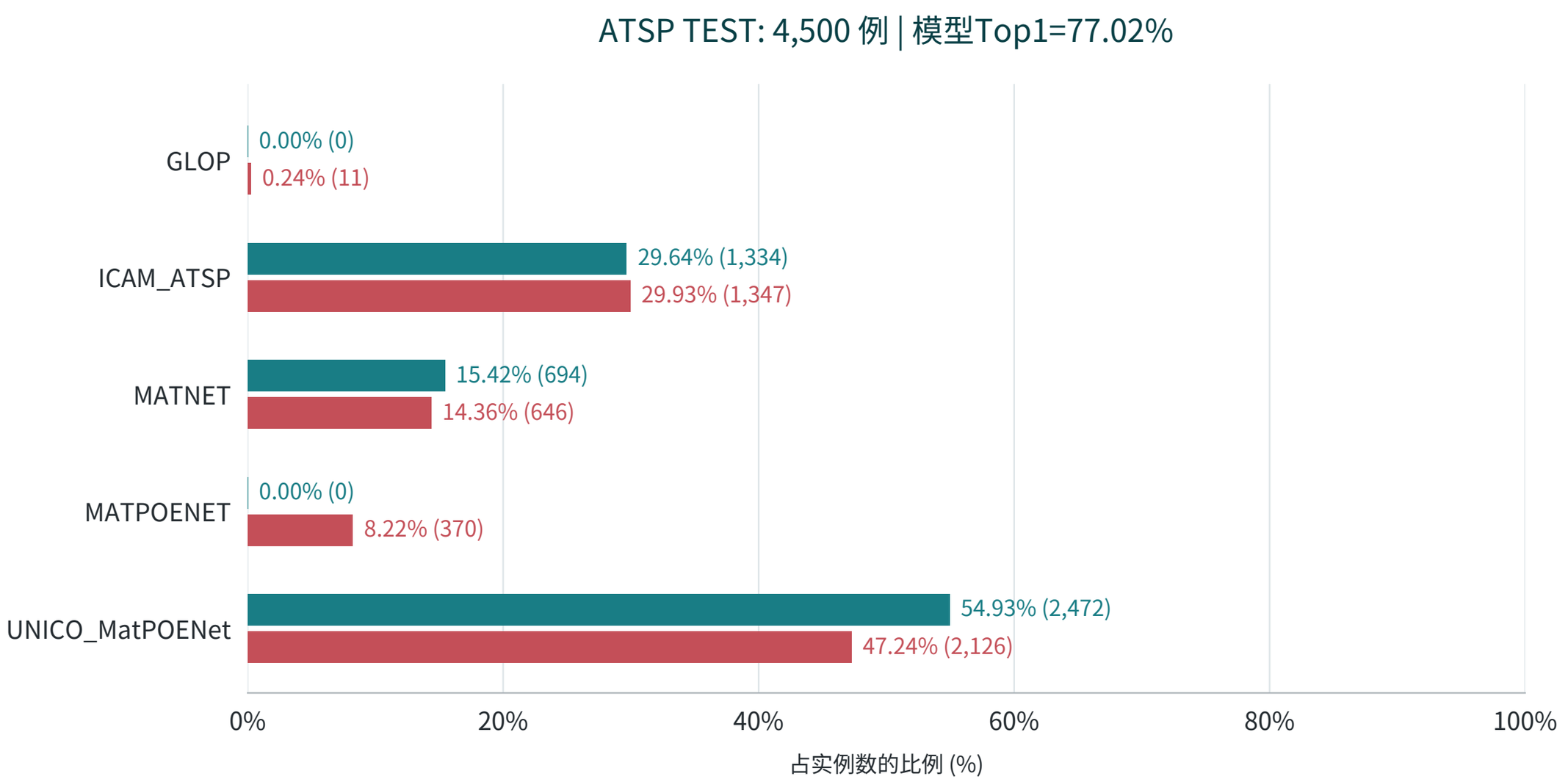
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | ATSP | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

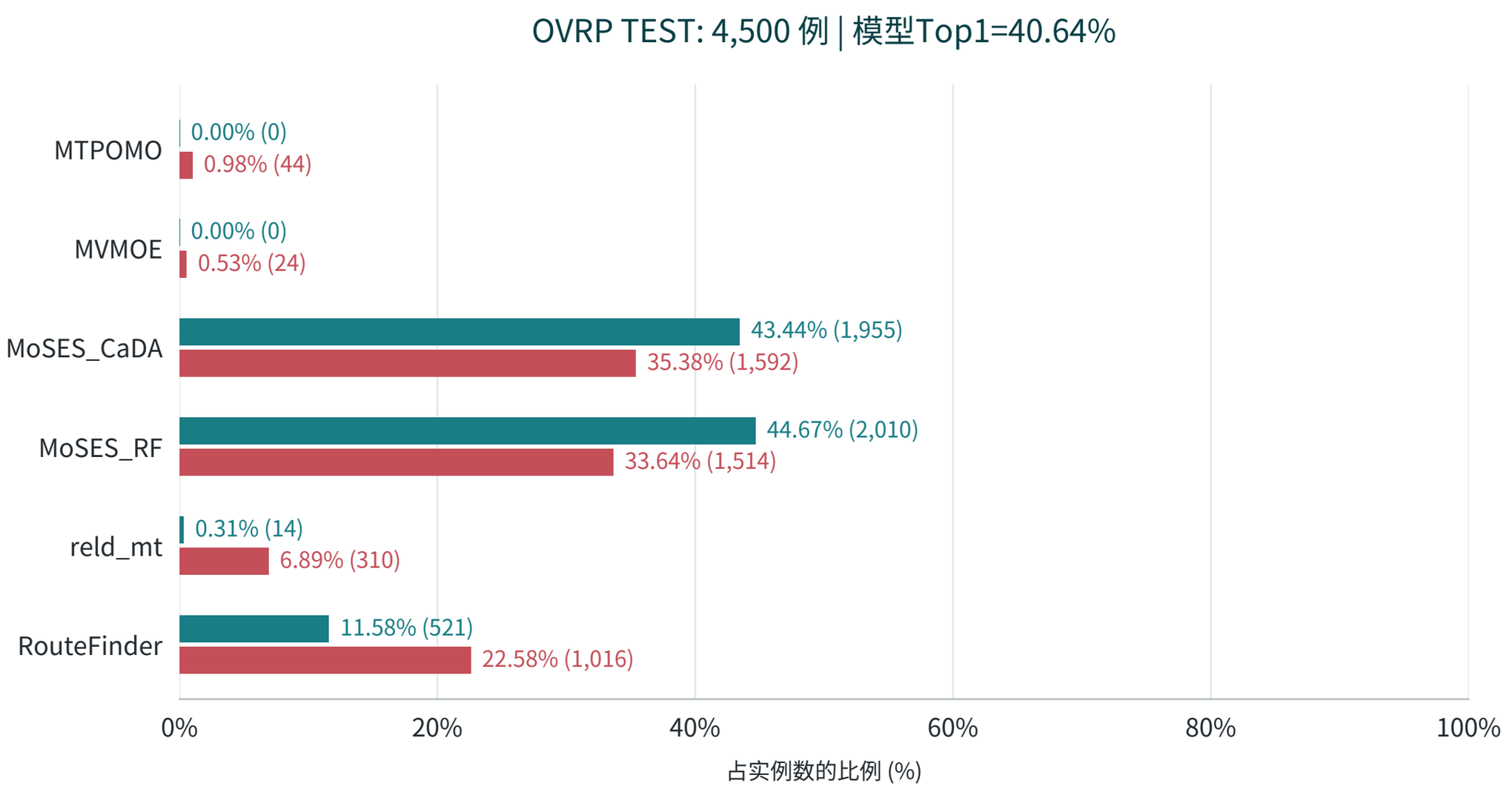
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRP | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

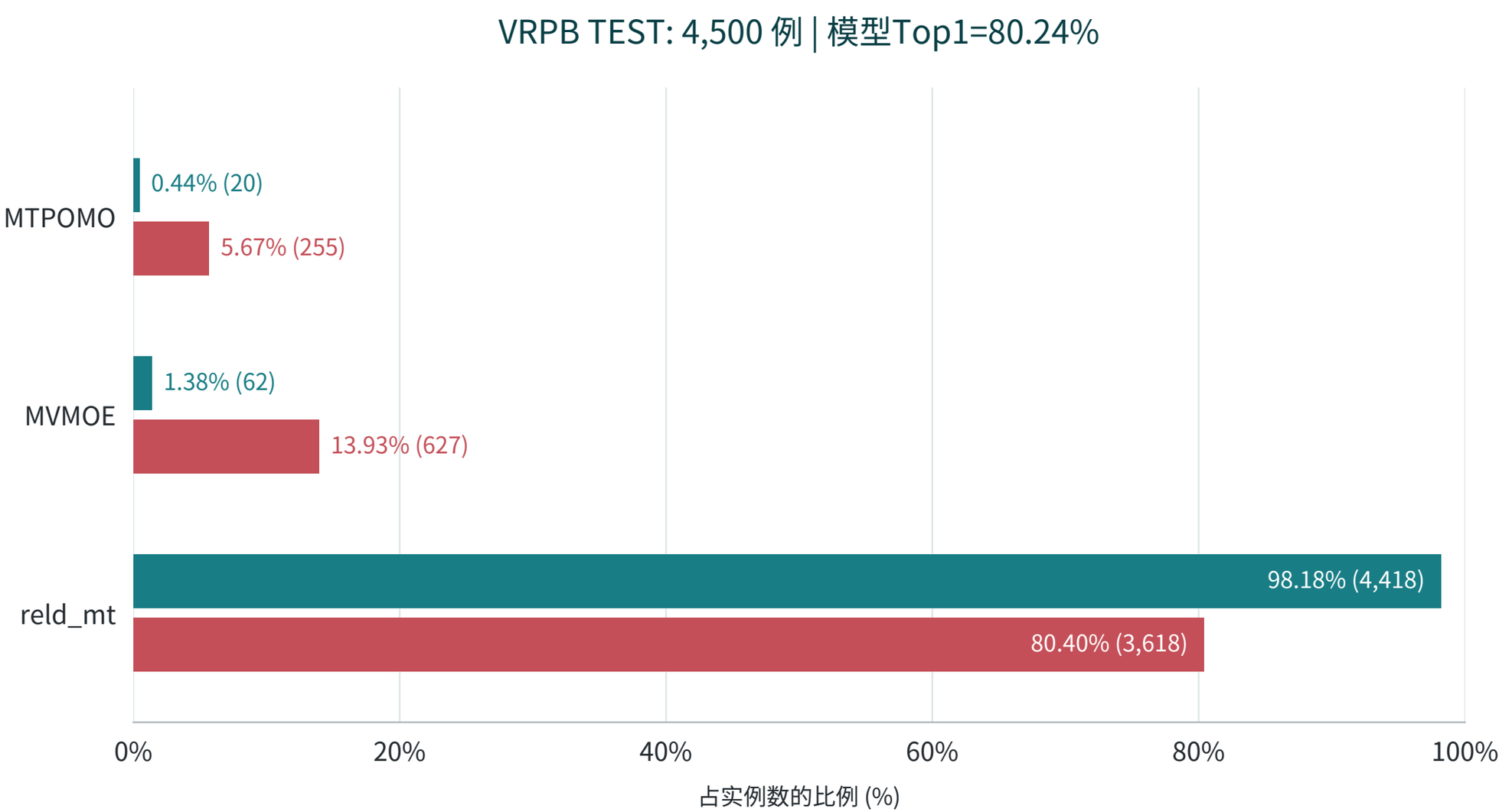
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPB | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

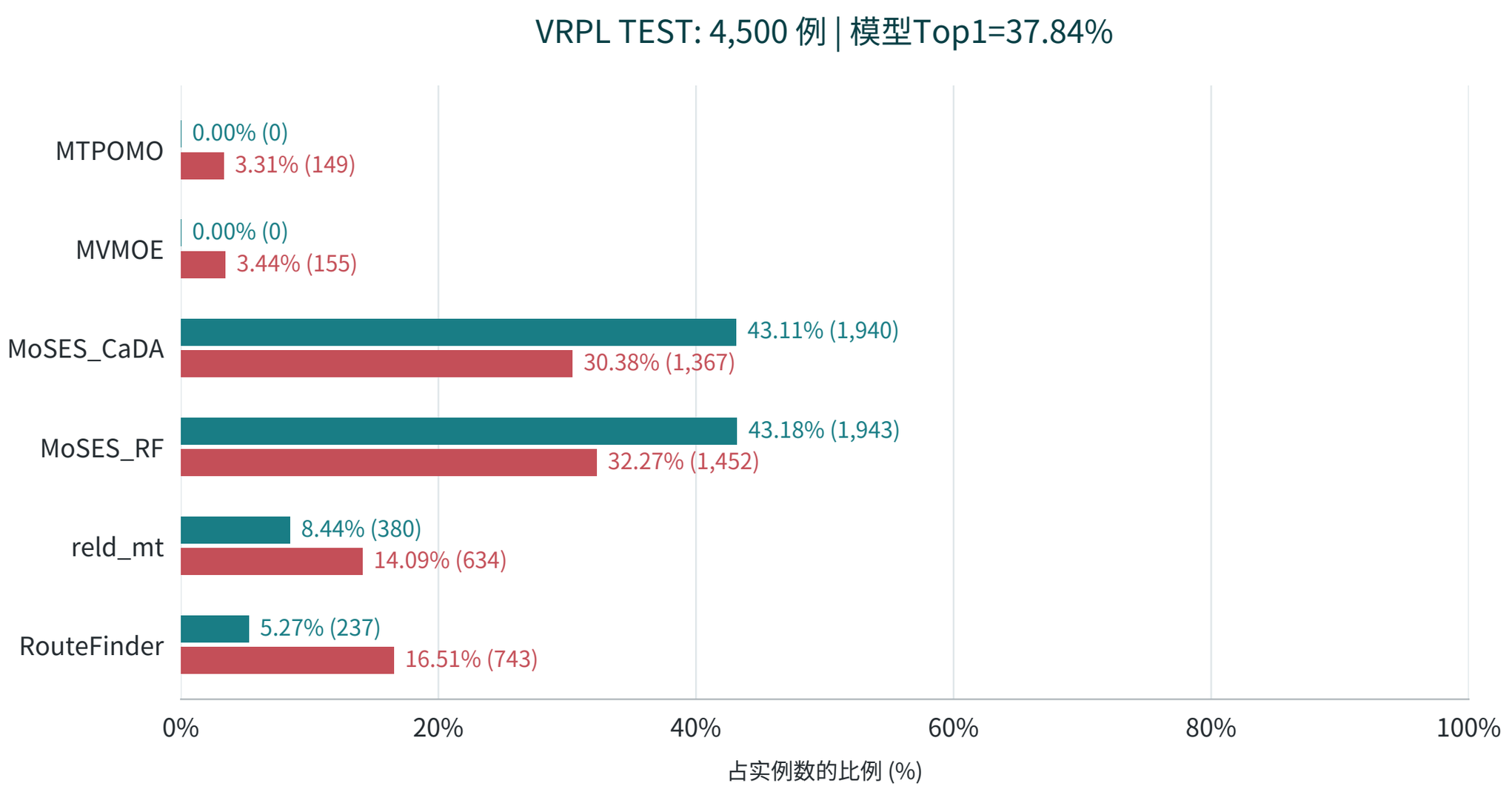
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPL | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

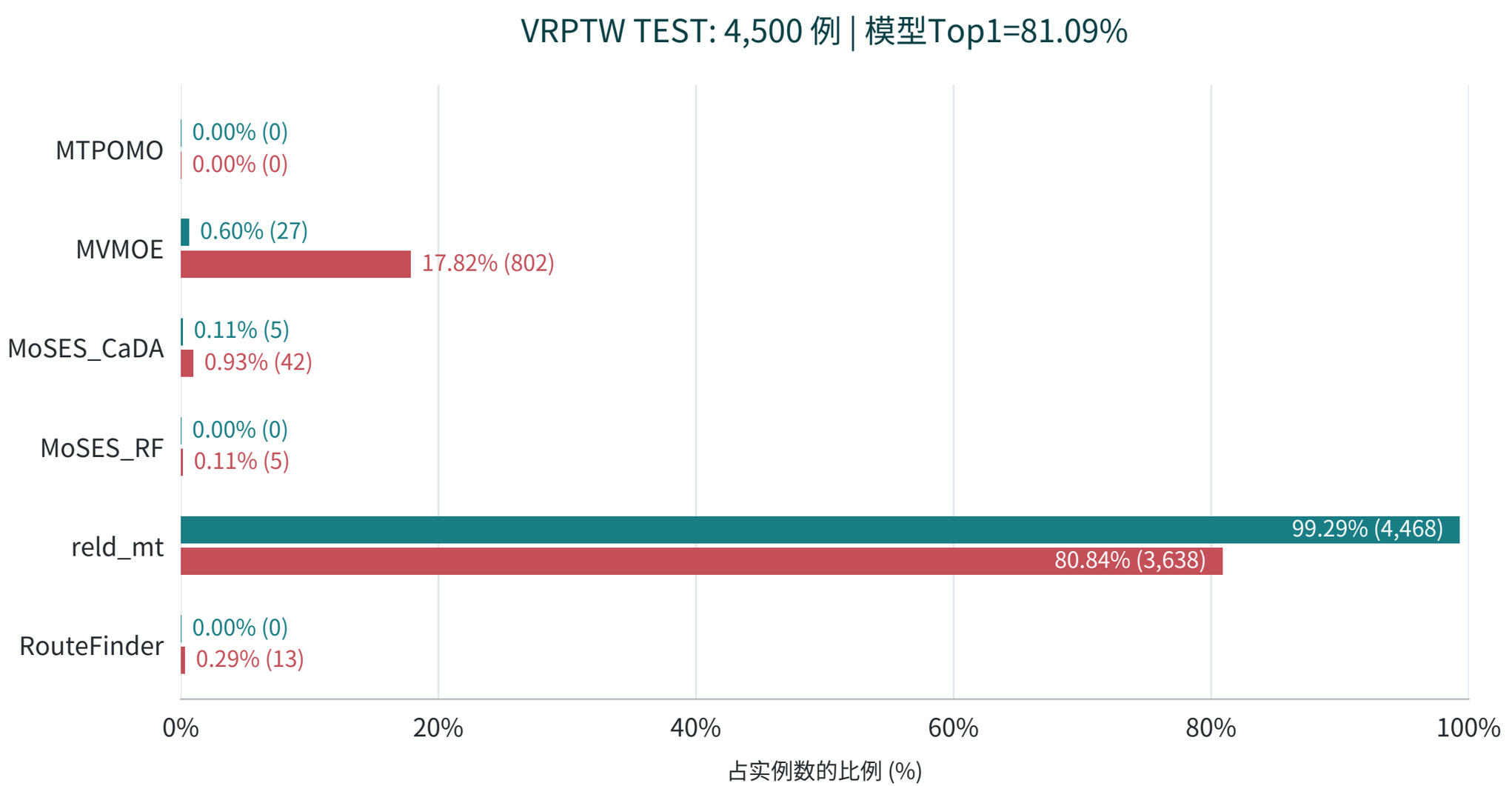
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPTW | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

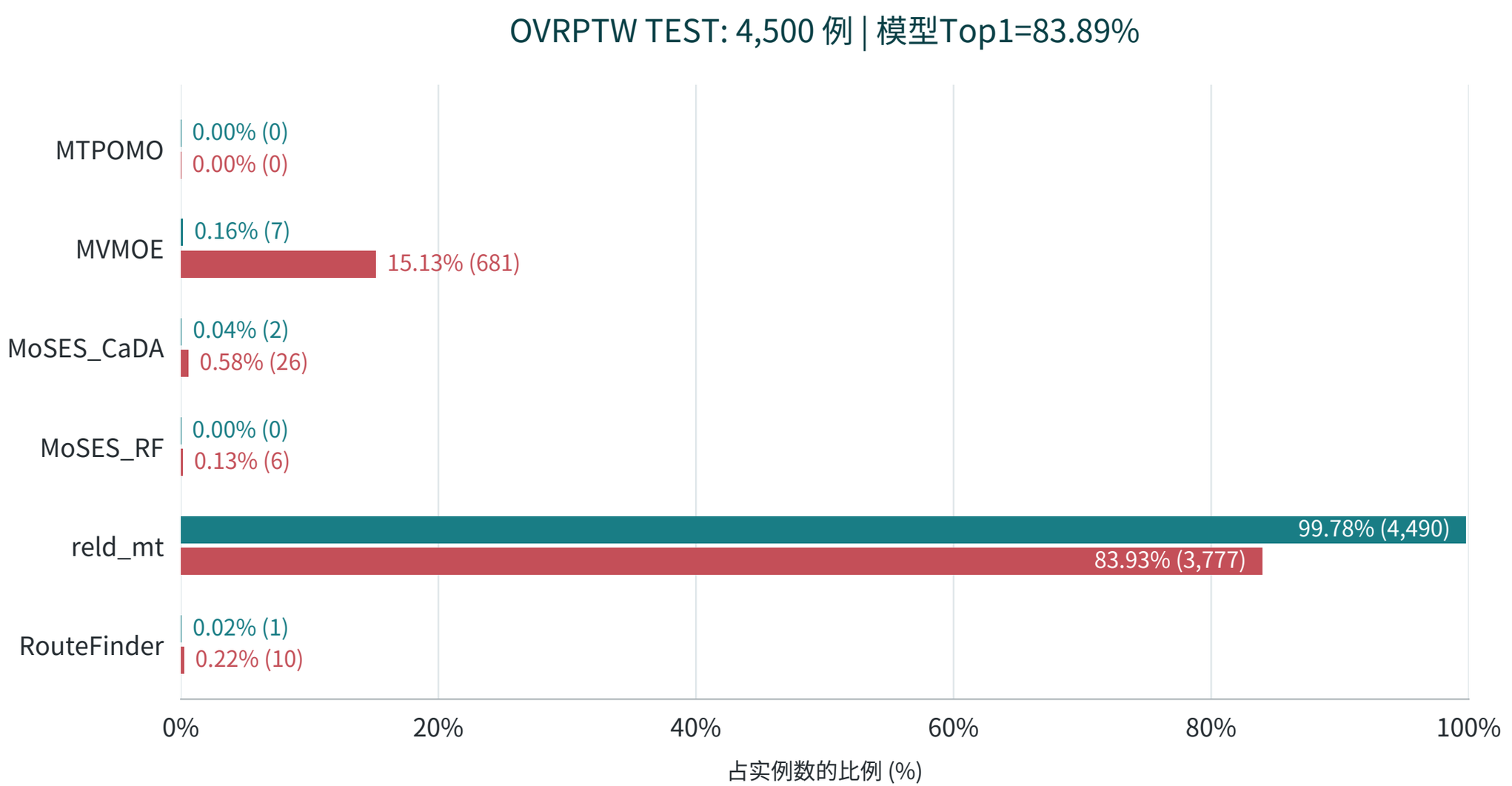
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPTW | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

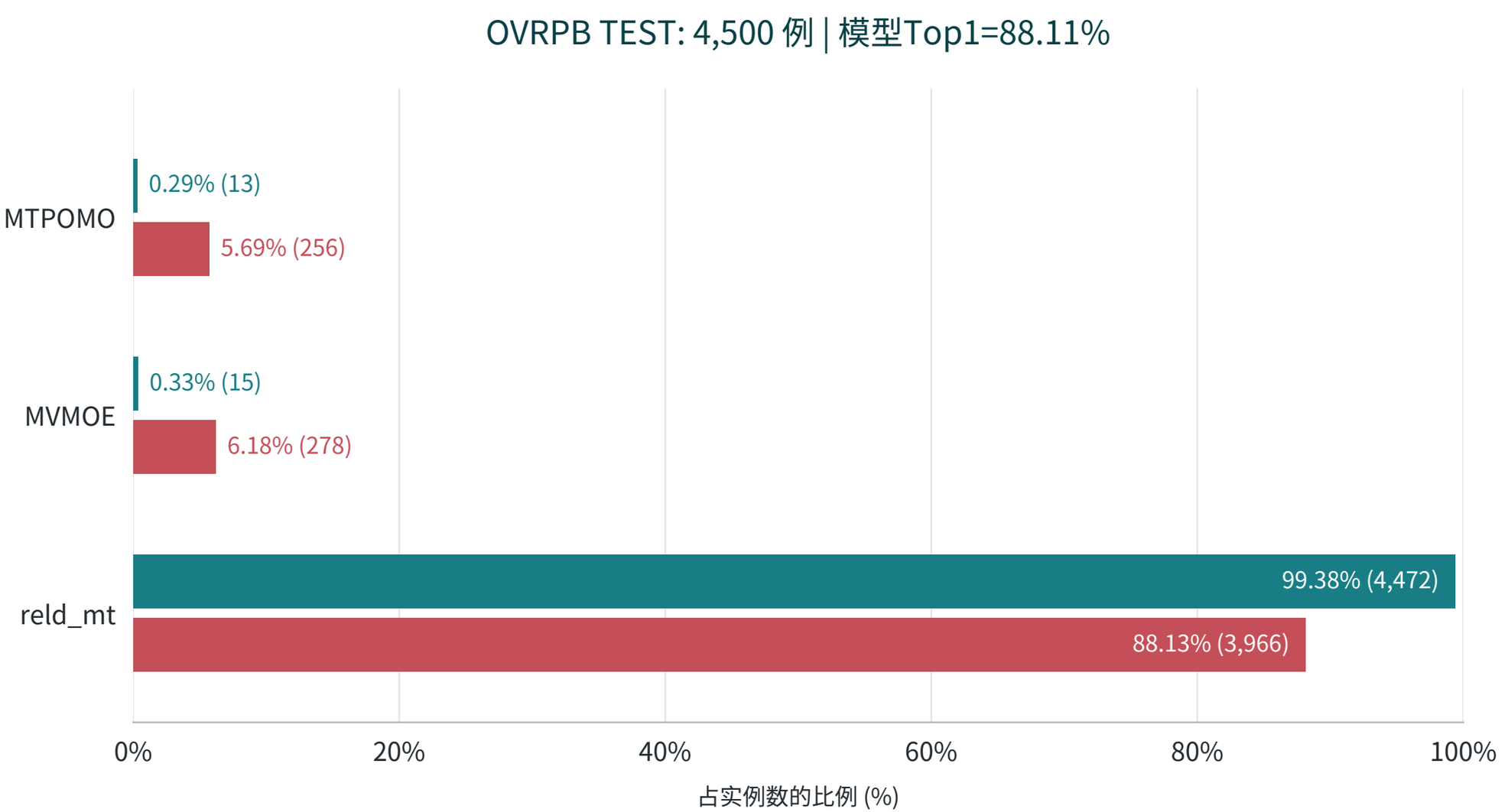
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPB | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

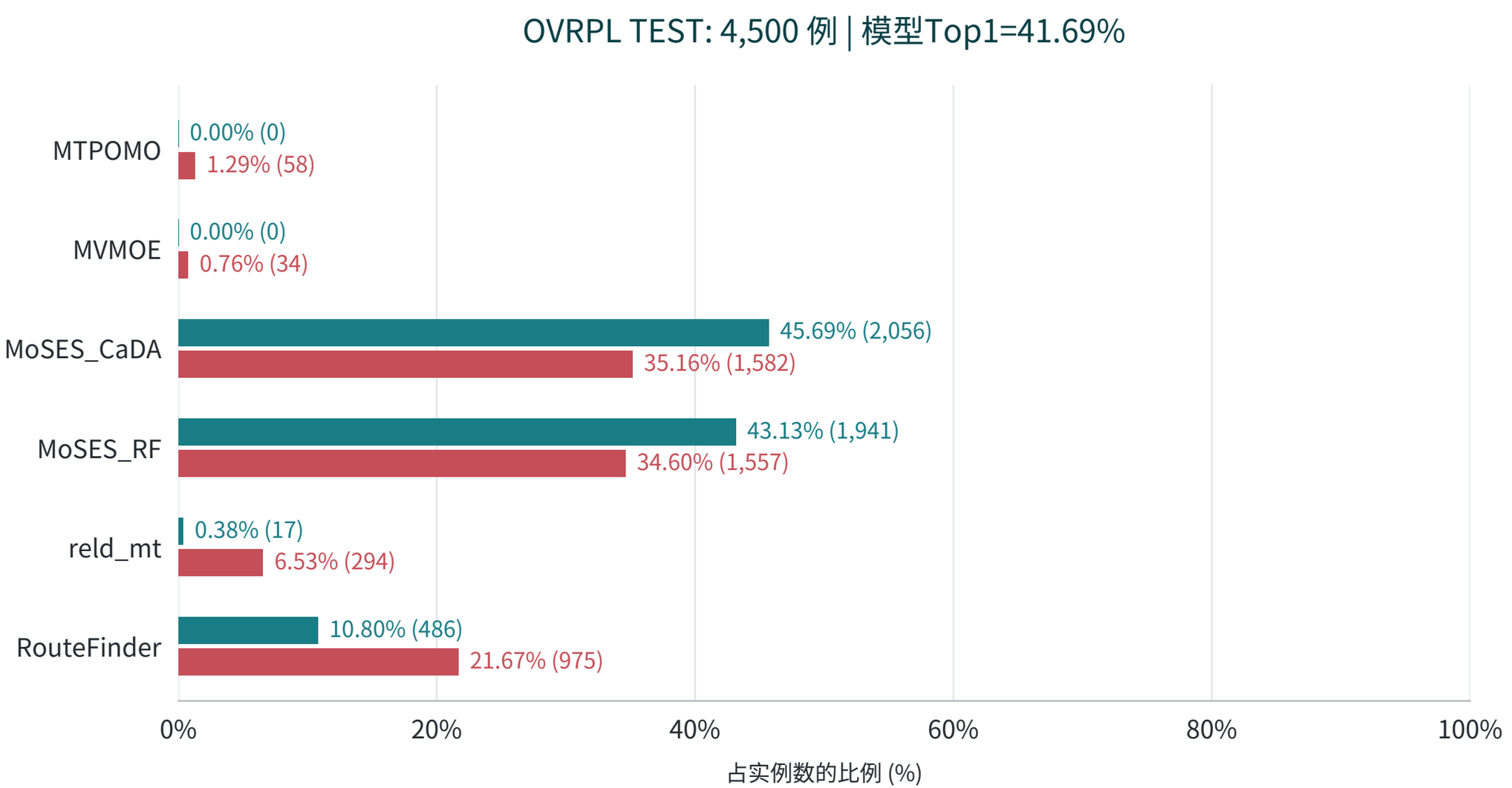
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPL | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

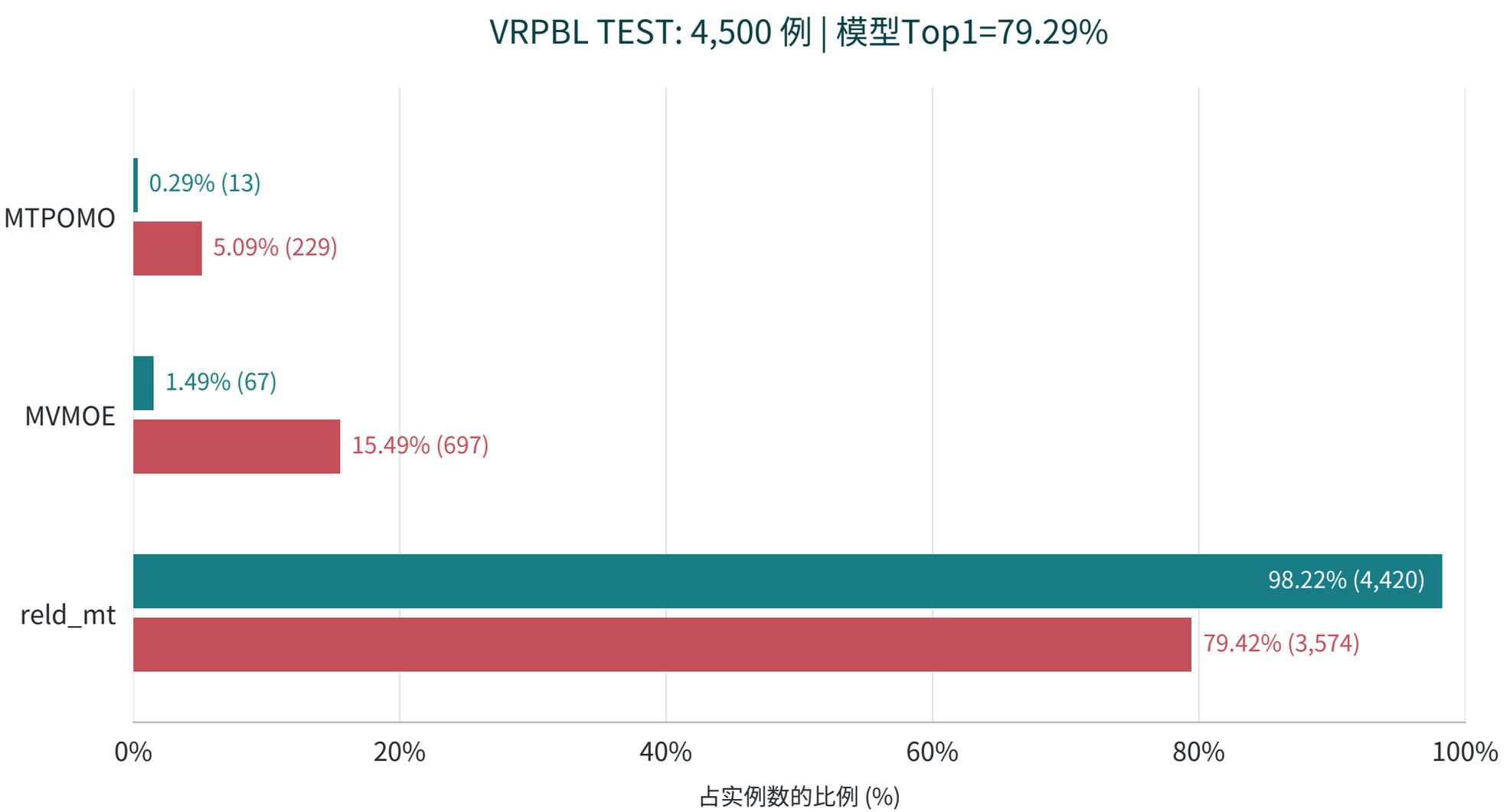
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPBL | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

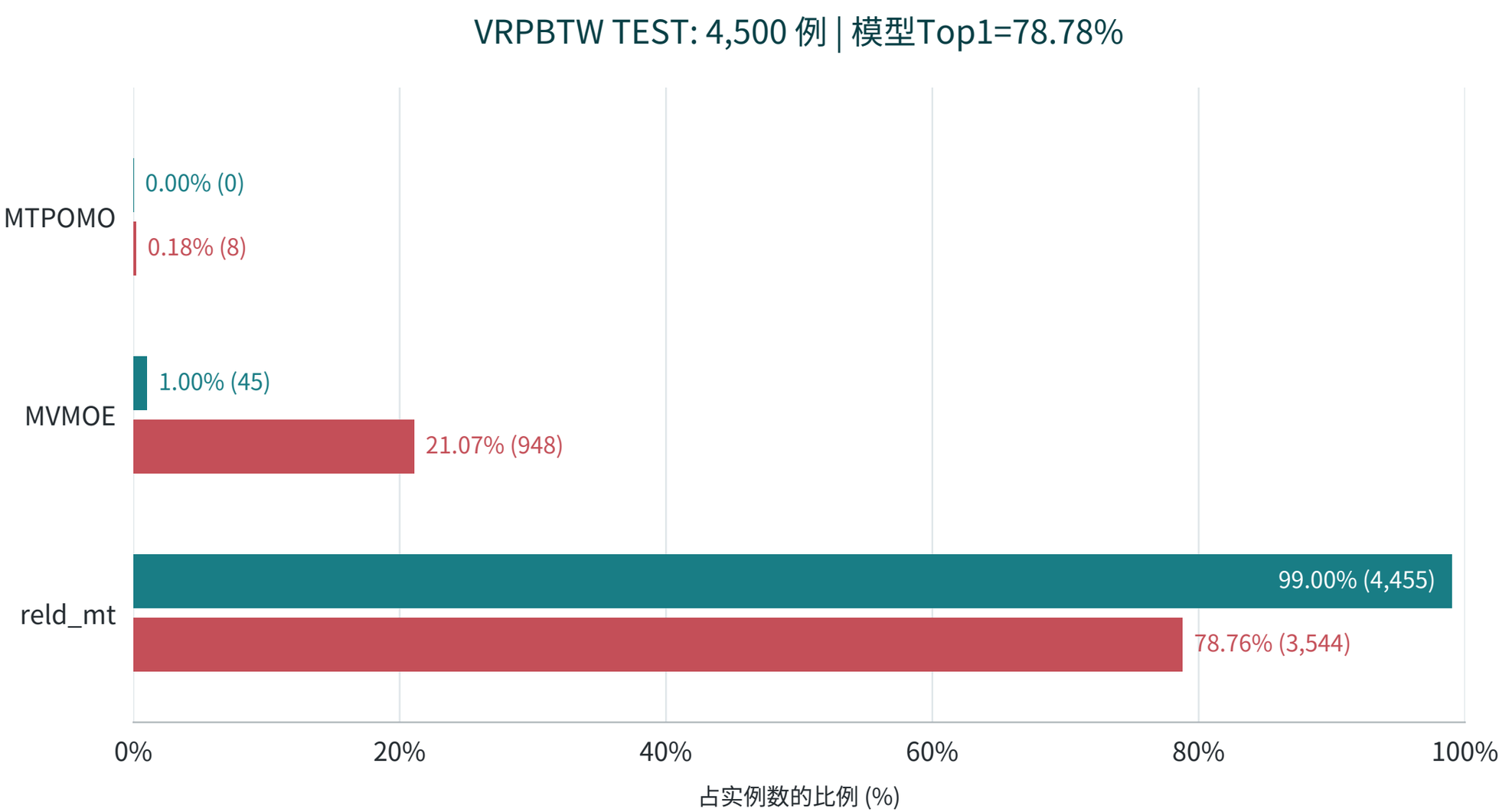
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPBTW | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

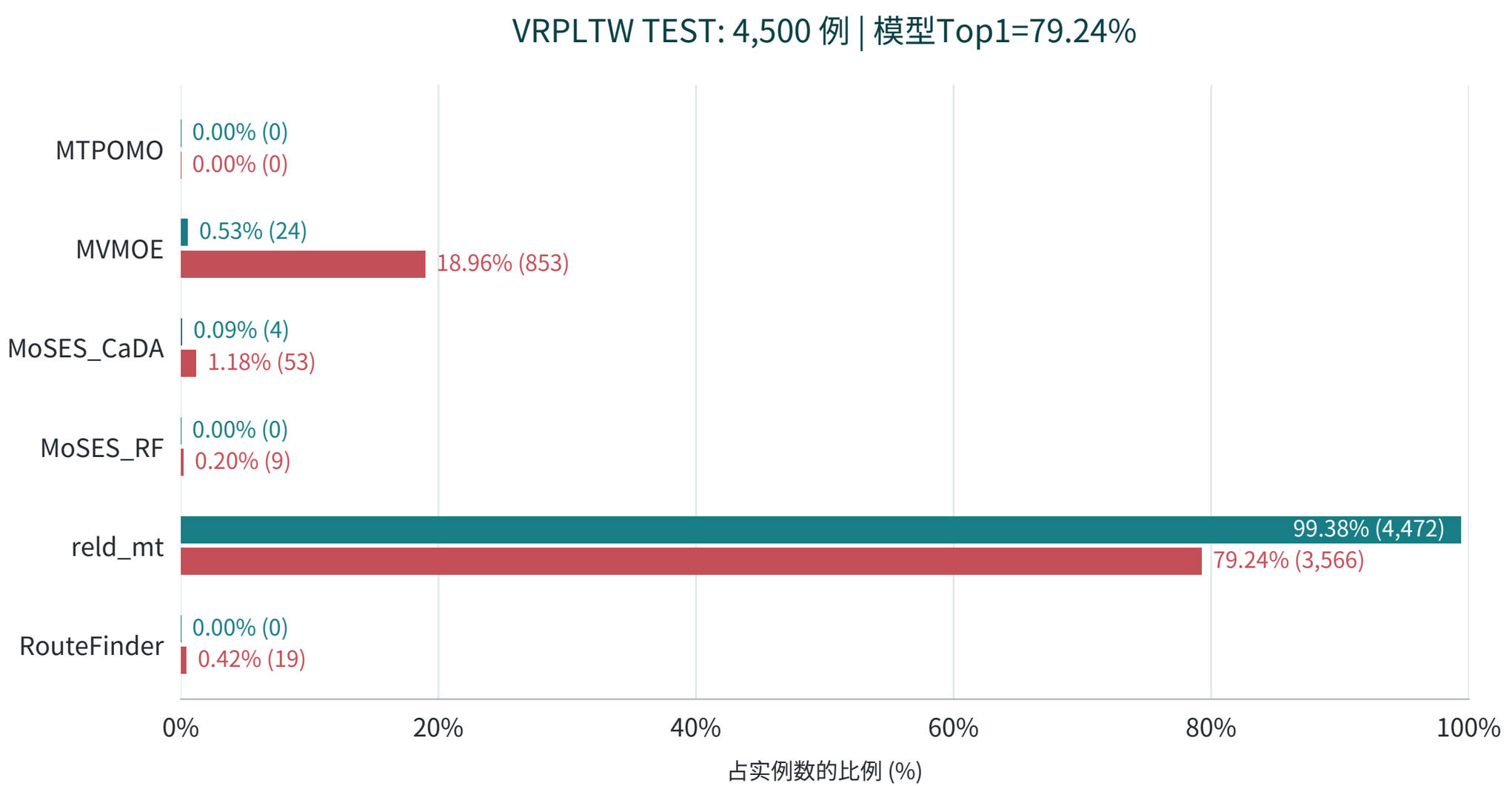
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | VRPLTW | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

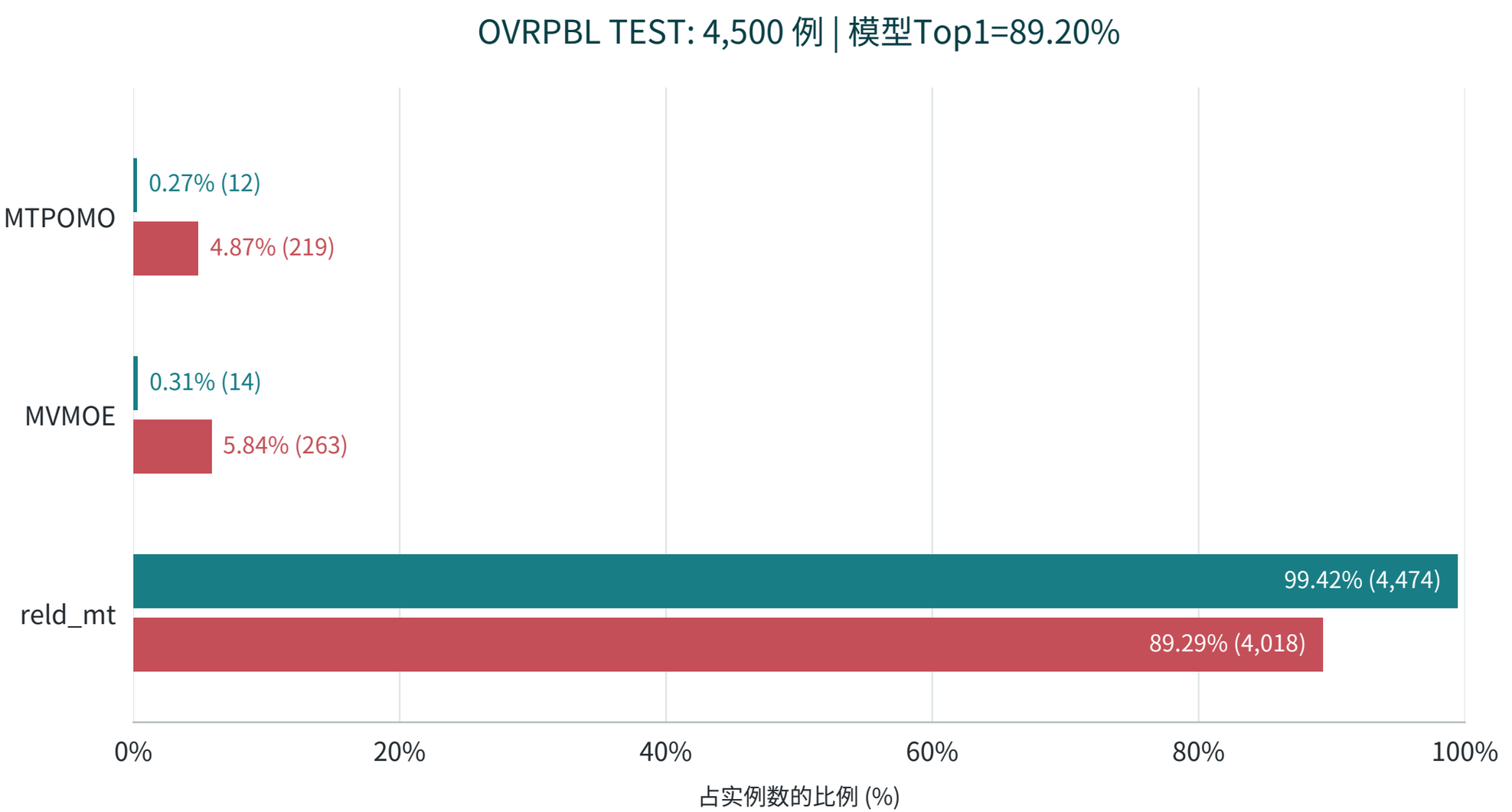
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPBL | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测




标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

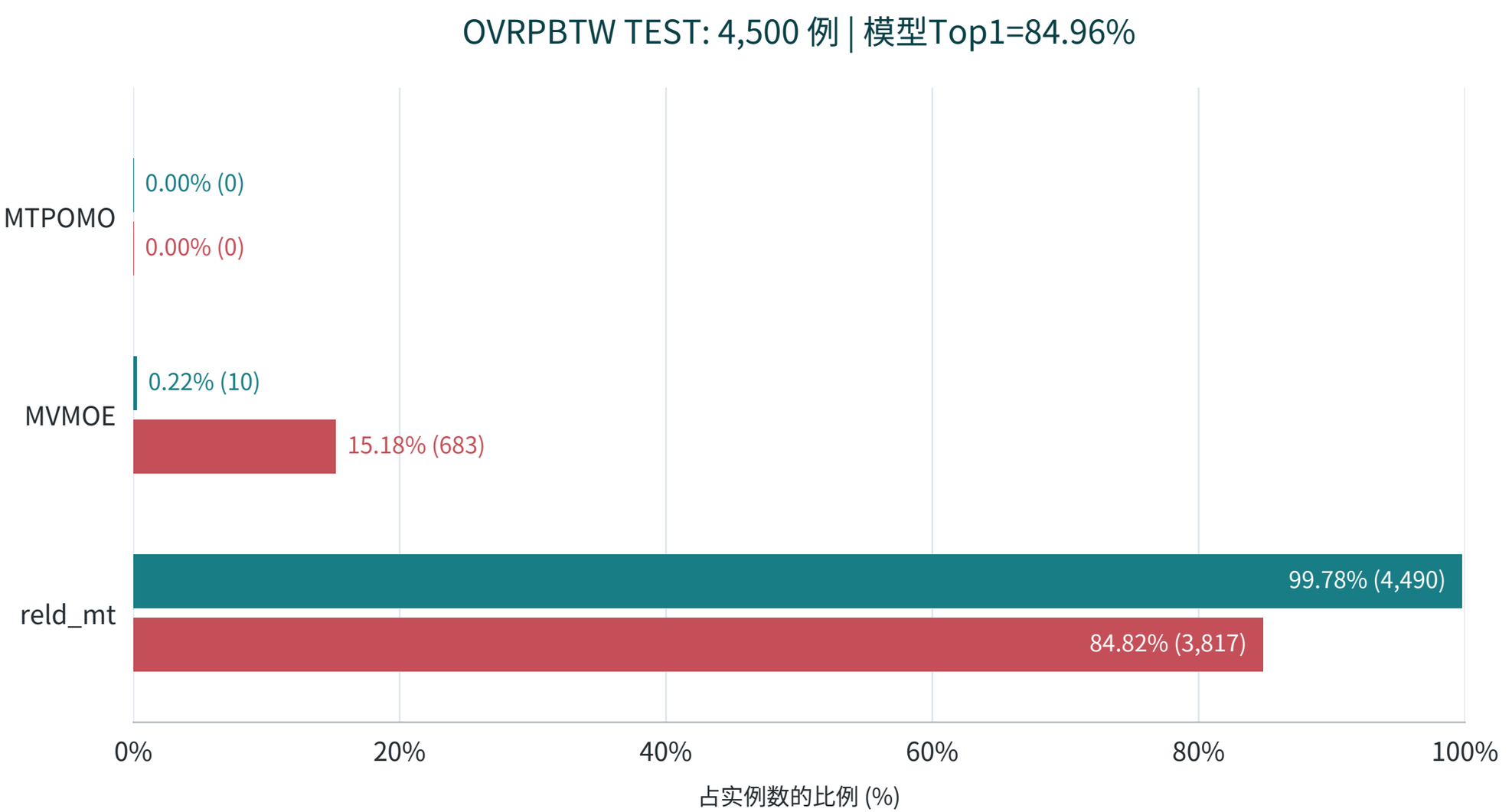
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPBTW | 方法分布

同一锁定模型: epoch 20 / update 57600 / SHA256 4bf692f36fde2f3f

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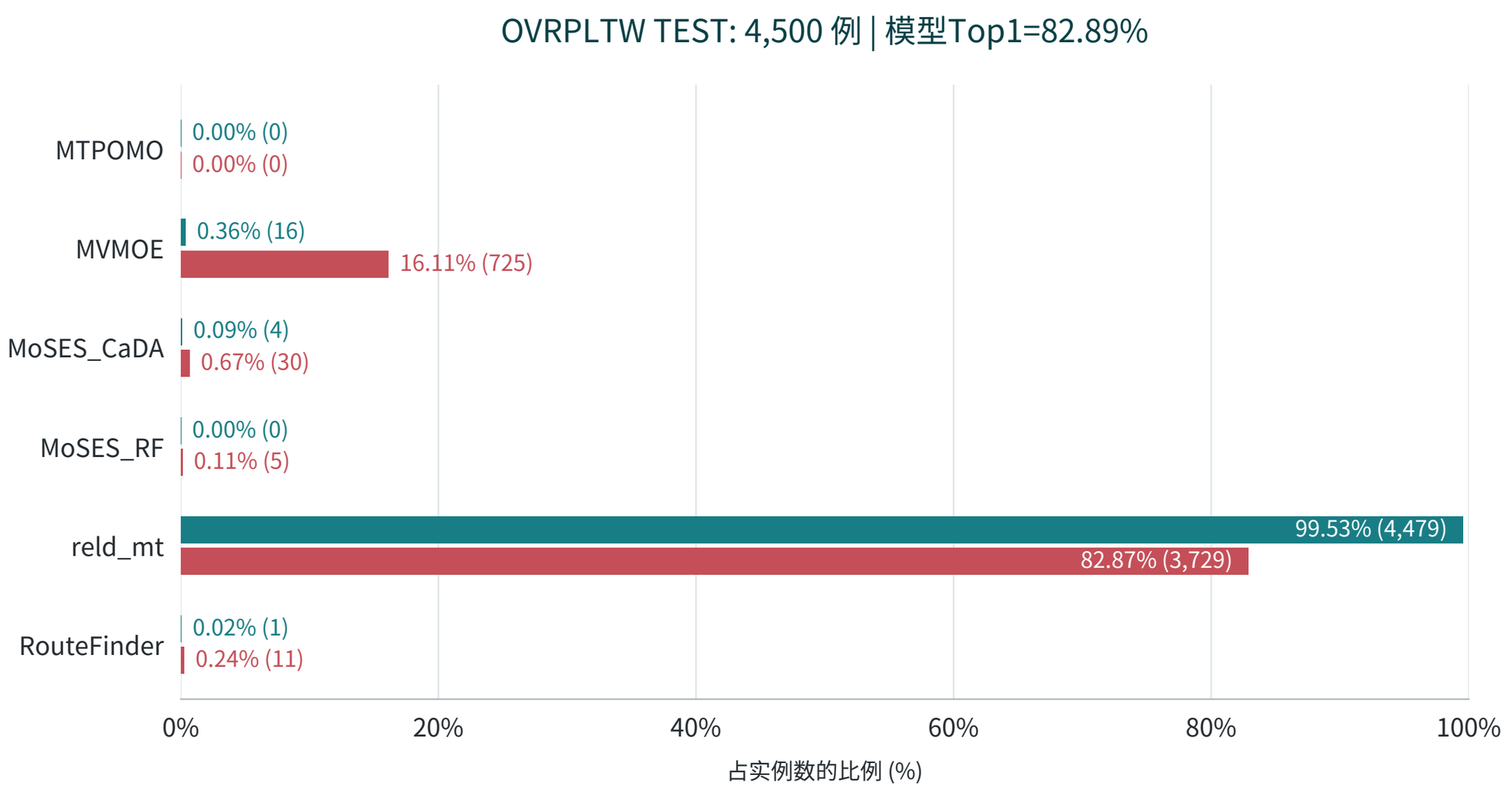
R112/A_collapsed: stable global solver-ID score ties. Same R112 lock; LIB inference only; no training/reselection/Rxx mixing.

# R112/A_collapsed | OVRPLTW | 方法分布

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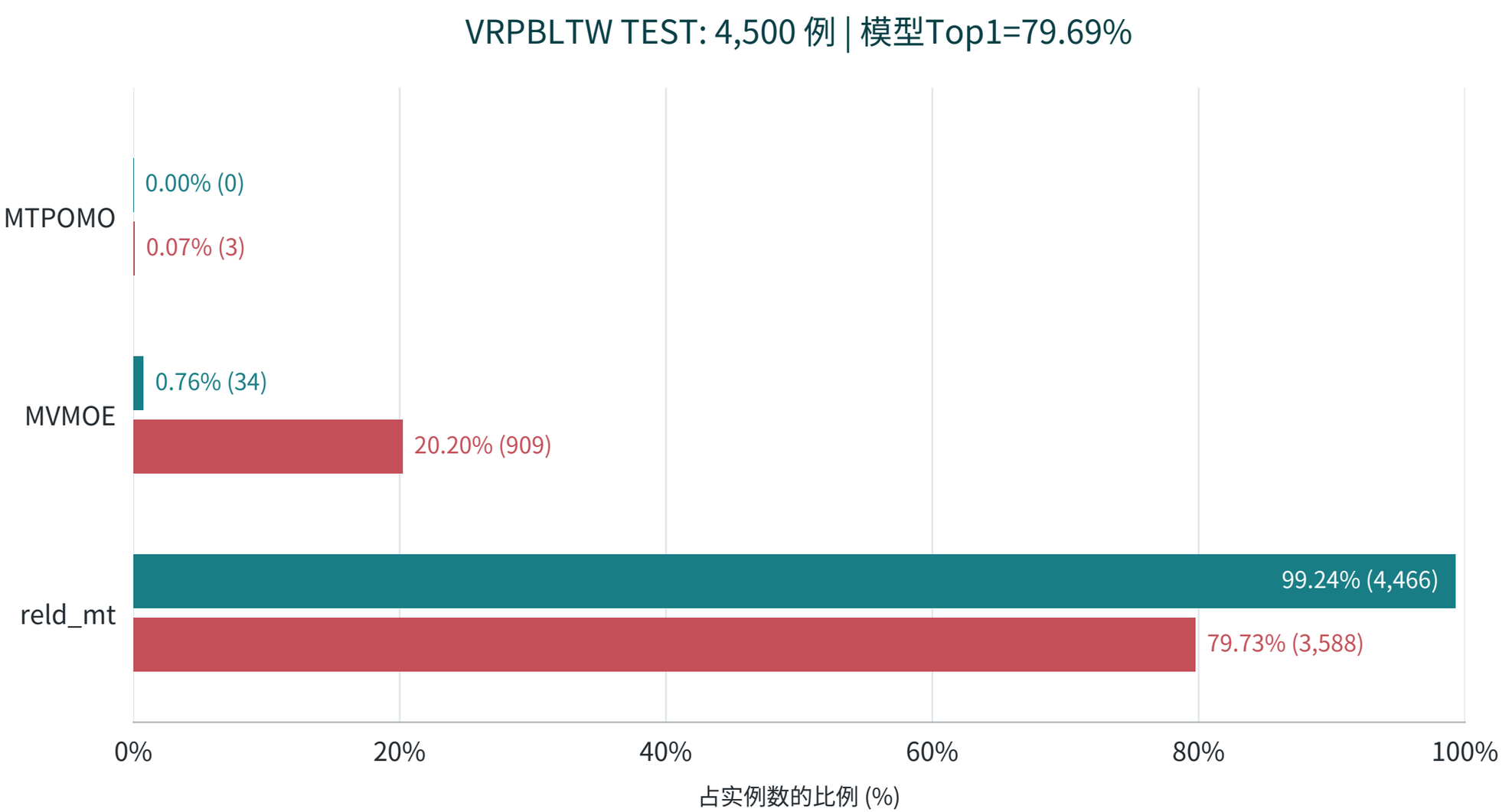
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# R112/A_collapsed | VRPBLTW | 方法分布

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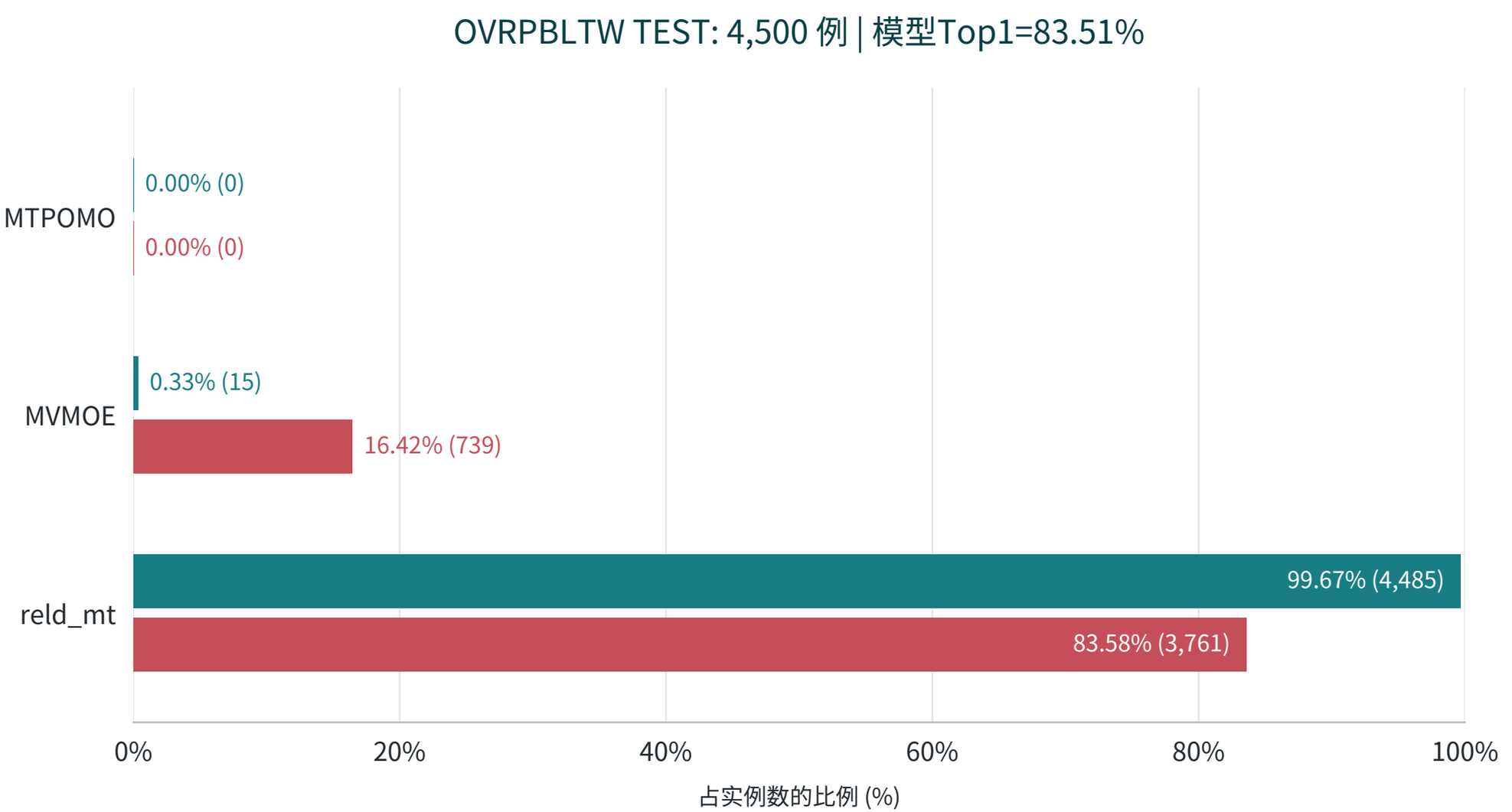
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